A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Biogeochemical cycles: carbon, nitrogen, phosphorus - Set 1

Mark scheme

AQA spec 3.2.4; The physical environment; textbook pp. 148-157

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 7 x 10^{11} tonnes **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 10% of 10 000 = 1000 billion tonnes **1** 1000 / 700 x 100 = 143% (accept 142.9) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) **1** Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** CO2 dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis **1** Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons **1** When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.5	**1 mark each** 1 mark for each of the 2 points below. **1** Aerobic respiration uses oxygen to break down organic compounds completely to CO_2 (and water), releasing much more energy **1** Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH_4), which is later oxidised to CO_2 in the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Ploughing:** increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster – carbon in the soil DOM store falls and atmospheric CO_2 rises **2 Draining:** waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly – drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO_2 (though methane release stops) **2 Marine pollution:** kills phytoplankton so less dissolved CO_2 is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton – less CO_2 then dissolves from the atmosphere to replace it, so atmospheric CO_2 rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air	6
02.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Rice paddy fields **1** Landfill sites **1** Anaerobic sediments in reservoirs **1** Livestock intestines (cattle) **1** Manure / slurry stores **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3

Do not write outside the box

Question	Marking guidance	Marks
02.3	1 mark for the point below. **1** More CO_2 dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions – more H+ lowers the pH **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Trees absorb CO_2 by photosynthesis and store carbon in wood as they grow **1** Once mature there is no further net storage (photosynthesis balanced by respiration / decomposition), though the standing forest remains a reservoir **1** Harvesting the timber and keeping it as wood (eg house construction) then replanting produces additional storage as the new trees grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
03.2	**1 mark each** 1 mark for each of the 4 points below. **1 Pre-combustion:** the fuel is modified before use so no CO_2 is released when burnt - eg gasification of coal to hydrogen and CO_2 (CO_2 stored, hydrogen distributed) – oxy-fuel combustion in pure oxygen giving only CO_2 and water vapour which is condensed out **1 Advantage:** capture at a large central facility – suits small dispersed sources (vehicles) that cannot capture CO_2 **1 Post-combustion:** CO_2 removed from exhaust gases by solvents, membrane filtration, adsorption / desorption, cryogenic separation, graphene **1 Disadvantage:** only about 20% of flue gas is CO_2 so separation is costly / energy-intensive – none fully developed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground • **adsorption** → nutrient ions sticking to the surfaces of clay and humus	4

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Injected into depleted oil / gas fields or saline aquifers **1** Injected into oil reservoirs to provide pressure for secondary oil recovery **1** Reaction with fractured basalt to form solid carbonate minerals **1 Concern:** leakage back to the atmosphere / monitoring over centuries – limited suitable geology – cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 151-152; AQA spec 3.2.4.2.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** 1 nitrogen fixation – 2 nitrification **1** 3 denitrification – 4 ammonification **1** 5 ionisation – 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **nitrification** → ammonium oxidised to nitrite and then nitrate • **denitrification** → nitrate converted back to nitrogen gas	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Drainage makes the soil aerobic **1** Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate **1** Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **nitrifying** → ammonium oxidised to nitrite and then nitrate • **denitrifying** → nitrate converted back to nitrogen gas	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrogen gas is very unreactive so most organisms cannot use it directly **1** Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate – the cycle recycles a small usable amount **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins **1** When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrification** → ammonium oxidised to nitrite and then nitrate	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Human = 120 + 60 + 30 = 210 – total = 325 **1** 210 / 325 x 100 = 64.6% (accept 65%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • A bare wrong answer with no working.	2
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Nitrogen and hydrogen react to form ammonia over an iron catalyst at high temperature and pressure – ammonia then converted to nitrates for fertiliser **1** Uses large amounts of energy from fossil fuels (also the hydrogen source), releasing CO2 and NOx **1** Biological fixation by bacteria occurs at low temperature using solar energy captured by plants (enzymes lower activation energy) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	3

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Question	Marking guidance	Marks
05.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Eutrophication of water bodies by leached nitrates / sewage **1** Photochemical smogs and tropospheric ozone from NOx **1** Acid rain (nitric acid) **1** Nitrous oxide as a greenhouse gas **1** NOx toxicity / respiratory disease **1** Nitrate in drinking water (methaemoglobinaemia) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Eutrophication** → nutrient enrichment of water leading to algal growth • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Carbon is lost as CO2 (composting) or CH4 (digestion) so the C:N ratio improves / more nitrogen-rich **1** Bulk and odour reduced, pathogens killed, easier to apply – digestion also yields biogas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 154-155; AQA spec 3.2.4.3.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) **1** Most phosphorus compounds have low solubility so little is available in soil water **1** Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

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Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil **1** Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mycorrhizal** → a fungus living with plant roots that improves nutrient uptake • **symbiotic** → a close relationship in which both species benefit	2
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) **1** Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment **1** Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates) – harvesting removes phosphorus from farmland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
06.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock **1** Breed crops that absorb phosphate more efficiently **1** Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) **1** Recover phosphate from sewage effluent by iron(III) sulfate precipitation – reduce soil erosion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **mycorrhizal** → a fungus living with plant roots that improves nutrient uptake **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	

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Question	Marking guidance	Marks
07.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Carbon:** combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere – deforestation reducing photosynthesis – ploughing / drainage increasing aerobic decomposition – new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane – marine pollution reducing phytoplankton – consequences: rising CO2 and CH4, climate change, ocean acidification, reduced soil organic matter • **Nitrogen:** Haber process doubling fixation – legume cultivation – drainage increasing nitrification and reducing denitrification – combustion NOx – sewage and fertiliser leaching – consequences: eutrophication, smogs, acid rain, N2O warming, NOx toxicity • **Counter-strategies carbon:** renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) • **Counter-strategies nitrogen:** reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones • **Evaluation:** some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS) – scale of fossil fuel dependence – feedback and time lags – link to circular economy and low-temperature natural processes – judgement	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **nitrification** → ammonium oxidised to nitrite and then nitrate • **denitrification** → nitrate converted back to nitrogen gas • **leaching** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (8 marks)	
	Humans have shifted the carbon cycle by burning fossil fuels and biomass, moving carbon from long-term stores into the atmosphere faster than photosynthesis and ocean uptake remove it, while deforestation reduces photosynthesis, ploughing and drainage speed aerobic decomposition, and paddies, landfills and livestock create new anaerobic sources of methane. The nitrogen cycle has changed as much: the Haber process has roughly doubled fixation, legume crops add more, drainage increases nitrification and reduces denitrification, combustion produces NOx, and leached fertiliser and sewage nitrate cause eutrophication, while nitrous oxide adds to warming.	
	Carbon can be counteracted by replacing fossil fuels with renewables and nuclear power, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, and increasing soil organic matter through low tillage and mulching, with carbon capture and storage as an unproven option. For nitrogen, catalytic converters and urea sprays reduce combustion NOx, biological fixation by legumes and rotation can replace manufactured nitrate, biological wastes can be managed as fertiliser, and low-solubility fertilisers, careful timing and buffer strips reduce leaching.	
	Some strategies are cheap and established - rotation, buffer strips, fertiliser timing - but others are expensive or untested, and the scale of fossil-fuel dependence, together with feedback and time lags, means the atmosphere will respond slowly. The strategies that mimic natural low-temperature cycling, as in the circular economy, are the most sustainable; the most effective is simply to leave carbon in the ground.	
	Weaker answer - Level 1 (3 marks)	
	Humans have changed the carbon cycle by burning fossil fuels which releases carbon dioxide, and by cutting down trees. They have changed the nitrogen cycle by using fertilisers which run off into rivers and cause eutrophication. To counteract this we could use renewable energy instead of fossil fuels and plant more trees. We could use less fertiliser. These would help the cycles go back to normal.	
	What makes the difference	
	• Level 1 names one or two changes per cycle; Level 3 covers several processes for each (decomposition, methane sources, Haber process, nitrification, NOx).	

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Question	Marking guidance	Marks
	• Level 3 connects each counter-strategy to the process it reverses; Level 1's strategies are general.	
	• The evaluation in Level 3 distinguishes established from unproven strategies and comments on scale and time lags - Level 1 has no evaluation.	
07.2	1 mark for the point below. **1** The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	
08.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (120 + 95) - (2 + 45) = 168 t ha^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO2 **1** Harvest removes biomass so less dead organic matter is returned – pasture roots and litter constantly add organic matter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Low / zero tillage **1** Adding manure, compost or mulch **1** Growing cover crops / leaving crop residues **1** Converting to long-term grass or tree crops in rotation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	

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